

Krishi Pracharak — Solution Document

Team

Name	Role
Rohit Raj	Architecture, AI Pipeline
Shivam Kumar	Backend, Product Design
Nisha Singh	Frontend, Streamlit Workflow
Alok Sharma	Data Collection, Analysis

GitHub: <https://github.com/Nishu3005/Krishi-pracharak>

Problem Statement

Syngenta field managers spend hours manually identifying which farmers to target, writing campaign messages in multiple languages, and briefing field reps — all from raw spreadsheets. The result is slow campaigns, generic messaging, and missed timing windows tied to crop growth stages.

Core challenge: How do you turn a dataset of 6,000 farmer profiles into personalised, timely, multilingual campaign actions — at scale, with no manual effort?

Our Solution — Krishi Pracharak ("Farm Advocate")

An AI-native agricultural marketing platform that takes a crop, a product, and a target region — and automatically produces:

- 1. A **ranked list of receptive farmers** scored by 7 data signals
- 2. **Micro-segments** clustered by crop × state × channel × persona
- 3. **Multilingual campaign content** — WhatsApp, SMS, IVR, poster — in Hindi, Punjabi, Marathi, Gujarati, Kannada, Bengali
- 4. A **field rep briefing** with ranked actions, restock alerts, and visit gap flags

Everything runs end-to-end from a single UI with no manual steps between data and deployment.

How AI / ML Adds Value

1. Receptivity Scoring Model (7 signals)

Every grower in the dataset is scored 0–1 before any campaign is sent. The score combines:

Signal	Weight	Logic
Timing urgency	+0.25	Peaks 7 days before a critical crop stage (sowing, heading, harvest)
State baseline	+0.15	Madhya Pradesh highest; Haryana lowest (calibrated from dataset)
Farm size tier	+0.10	1–2 acre farms respond best to digital campaigns
Scan signal	+0.06	Grower scanned a Syngenta product QR — strong buying-intent signal
Engagement history	+0.05	Previous WhatsApp message opened
Gender signal	+0.02	Female growers show documented uplift in this dataset
Fatigue penalty	−0.08	Contradictory device signals — likely stale data, deprioritised

Eligibility threshold: score ≥ 0.30. Below this, the grower is not targeted. This prevents wasted spend on unreceptive farmers.

2. AI Segmentation Agent (Claude Sonnet)

After scoring, the targeting agent clusters eligible growers by (crop × state × language × channel × persona). A minimum cluster size of 20 growers prevents micro-fragments. The agent then calls Claude Sonnet once to produce a campaign rationale — explaining which segments to prioritise and why, grounded in the score distribution and crop calendar.

3. Content Generation Agent (Claude Haiku)

One LLM call per segment (not per grower) produces all four content formats:

- **WhatsApp** — ≤200 chars, native language, persona-matched tone
- **SMS** — ≤120 chars, native language
- **IVR script** — 30-second spoken script
- **Poster image** — AI-generated via Gemini 2.5 Flash

Each call includes automatic retry and a safe fallback — no campaign fails silently.

4. OOS Gate (Out-of-Stock Intelligence)

Before any content is generated, the system checks inventory trends across 4,000 retailers. If a product is trending out-of-stock in a territory, that segment is blocked — preventing campaigns that drive demand for unavailable products.

5. Rep Briefing Agent (Claude Haiku)

For growers without smartphones (≈26% of the dataset), campaigns route to `rep_assist` instead of digital channels. The rep agent computes OOS slope from 4 weekly inventory snapshots, detects visit gaps (21-day threshold), and generates a ranked action briefing for each field rep.

Architecture

```

app.py — Home: Grower network + field force analytics
├── pages/1_campaign_builder.py
│   ├── agents/targeting_agent.py
│   │   ├── utils/receptivity_score.py
│   │   └── utils/crop_calendar.py
│   └── agents/content_agent.py
│       └── utils/ai_client.py (Token Router)
└── pages/2_rep_briefing.py
    └── agents/rep_agent.py

```

Agent coordination: `targeting_agent` writes `data/targeting_plan.json`. `rep_agent` reads it — agents share state via plain JSON, no message bus required.

No database. State flows through JSON files. Simple to inspect, simple to demo.

Key Features

Campaign Builder

- Create a campaign: crop, product, states, date window, channel mix
- AI targeting runs automatically — scores all 6,000 growers, segments, checks OOS
- 4 tabs per campaign: **Overview**, **Content Generation**, **Receptivity**, **Rep Briefing**

Overview Tab

- Dark campaign header with key metrics and pill badges
- Live weather cards for each active state (Open-Meteo, no API key needed)
- AI agent insight block — campaign rationale in plain language
- OOS warning block if any products are blocked
- Segment stats: scored growers, eligible count, offline growers, segments

Content Generation Tab

- Per-segment content scope table
- One click generates all formats for all segments
- WhatsApp + SMS side by side, IVR full width, poster with prompt
- Each variant shows status badge, character count, model used

Receptivity Tab

- Score distribution histogram with eligibility threshold line
- Per-signal contribution bar chart
- Open rate by state and farm size tier
- Channel eligibility split (smartphone vs non-smartphone)

Rep Briefing Tab

- Pick a rep and date → AI generates a ranked action plan
- Restock alerts, visit gaps, offline grower list
- Inventory intelligence: OOS rate by SKU, stock trends, risk heatmap

Home Dashboard (localhost:8501)

- Grower network: 6,000 farmers, 10 states, 33 districts
- Geography, crop, technology, and demographic analytics
- Full Syngenta field imagery hero with live metric strip

Models Used

Task	Model	Reason
Targeting rationale	claude-sonnet-4-6	Reasoning quality; called once per campaign
WhatsApp / SMS / IVR content	claude-haiku-4-5	High throughput, low cost; called per segment
Rep briefings	claude-haiku-4-5	Short output, low latency
Poster images	gemini-2.5-flash-image	Image generation; failure is non-blocking

All calls go through a single OpenAI-compatible client pointed at the Token Router base URL.

Dataset Coverage

Metric	Value
Total growers	6,000
States	10
Districts	33
Retailers tracked	4,000
Languages supported	6 (Hindi, Punjabi, Marathi, Gujarati, Kannada, Bengali)
Avg farm size	1.97 acres
Product scan rate	19.28%
Non-smartphone growers	~26%

Core Innovation

A crop-stage-aware, OOS-gated, multilingual AI campaign engine that scores every farmer in real time and delivers deployment-ready content and rep briefings in under two minutes — replacing days of manual field planning.

Tech Stack

Layer	Technology
Frontend	Streamlit + custom CSS (Space Grotesk + Source Serif 4)
Charts	Plotly
AI agents	Anthropic Claude (Sonnet 4.6 + Haiku 4.5) via Token Router
Image generation	Google Gemini 2.5 Flash
Weather	Open-Meteo (free, no key)
Storage	JSON files (no database)
Language	Python 3.11

Setup

```
git clone https://github.com/Nishu3005/Krishi-pracharak.git
cd Krishi-pracharak
pip install -r requirements.txt

# Create .env with:
# OPENAI_API_KEY=your_token_router_key
# OPENAI_BASE_URL=your_token_router_base_url

streamlit run app.py
```

Open <http://localhost:8501>